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Explained: India's 5G spectrum and rollout

Indian telecom operators are expected to begin 5G service offerings by Diwali 2022

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In the race to deploy 5G services in the country, Reliance Jio, Bharti Airtel, Vodafone Idea and Adani Data Networks are all intensely bidding for 72GHz of total spectrum allocation worth at least Rs 4.3 lakh crore to the Indian government.

If you have a 5G-enabled smartphone, and you want to understand some of the nitty-gritties of the 5G spectrum and various frequency bands involved, keep reading...

INDIA'S 5G SPECTRUM BANDS

In case you didn't know, the Indian 5G spectrum for auction includes 600MHz, 700MHz, 800MHz, 900MHz, 1800MHz, 2100MHz, 2300MHz, 2500MHz, 3300MHz, and 26GHz bands. These are nothing but wireless radio wave frequencies used for telecommunication services.

These Indian 5G spectrum bands are categorized as low, mid and high frequency bands based on their, well, frequency range. For example, the low band comprises all frequency ranges below 1GHz – 600MHz, 700MHz, 800MHz, and 900MHz. The high frequency bands fall between 26GHz to 40GHz in the mmWave range – great for high-bitrate data transfer over very short distances, one of the biggest promises of 5G to be honest.



Image by Freepik

The 4G networks were designed keeping in mind maximum smartphone use cases whereas 5G takes it to the next level by offering increased bandwidth and higher data speeds. Also, 5G supports several more use cases like deploying virtual networks, and software-defined networks and can connect to a variety of devices beyond just smartphones. In addition to streaming high-quality videos on the go and faster download speeds with 5G, a big push is also on how it will transform the gaming industry with cloud gaming and XR (Extended Reality) experiences.



WHY 700MHZ BAND MATTERS

On the first day of India's 5G spectrum auction, the 700MHz band of spectrum received bids for over Rs 39,000 crore by all four Indian telecom operators across all 22 circles of the country – that's close to 30-percent of all the bids submitted totalling over 1.4 lakh crore. In fact, by early estimated reports, the 700MHz band is proving to be the most expensive piece of 5G spectrum going under the hammer.

So why is this single band of 700MHz frequency getting so much attention from the likes of Reliance Jio and Bharti Airtel in India's 5G deployment race? According to experts, this 700MHz band is going to be crucial for wide-spread, uninterrupted 5G data transmission in the country. It's simple physics: radio waves with lower frequencies travel further through obstacles without degrading than waves with higher frequencies. Hence, the 700MHz band is being eyed by Indian 5G service providers with a lot of interest.